Declaration of AI assistance

Journal of Number Theory

Author-prepared disclosure; not an official publisher form or signature.

Manuscript. The optimal log log q bound in the Pólya-Vinogradov inequality

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Claude through Claude Code assisted the earlier coauthored manuscript, formal development and checking programs. OpenAI Codex assisted the explicit 11/12 parameter conversion, proof and scope auditing, Mellin arguments, the general smooth Mobius equivalence, new Lean analytic modules, verification records, editing and typesetting. OpenAI's original zero-free theorem remains a credited external mathematical contribution whose proof sources are supplied in the repository. This reproducibility revision adds a portable formal-source archive, fresh source and historical-report checks, a conditional mean-square target, a proof obstructing a coefficient-independent shortcut and exact parameter controls. The added research notes await author review.

Human oversight. The submitting author stated on 25 October 2026 that the earlier manuscript was a mix of human-written and AI-generated material with all parts checked and understood by a human. The added research material awaits author review. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The unchanged proof tree has the completed 19 original, seven endpoint and 20 component axiom reports, using only propext, Classical.choice and Quot.sound. Source identity and historical receipts were rechecked. No new full Lean build or independent zero-free proof is claimed; the new conditional research note is not formalized.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.elsevier.com/about/policies-and-standards/generative-ai-policies-for-journals

Prepared on 25 October 2026. The author reviews this statement and completes any additional live-portal declarations before submission.